

✓ Lab 4

```
import numpy as np
import matplotlib.pyplot as plt
import cv2
import requests
```

```
image_url="""
https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRiBWYFazCzI4ZamD6KuYXUuj8qtVaOMmNy4-V8oqhT7Q&s=10
"""
response = requests.get(image_url)
response.raise_for_status()
image_array_1 = np.frombuffer(response.content, np.uint8)
gray_1 = cv2.imdecode(image_array_1, cv2.IMREAD_GRAYSCALE)
```

```
image_url="""
https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcS77koaHnxnNHPIo2dL0mzv6CVVLI_Ay71AQYJ_fh_why&s=10
"""
response = requests.get(image_url)
response.raise_for_status()
image_array_2 = np.frombuffer(response.content, np.uint8)
gray_2 = cv2.imdecode(image_array_2, cv2.IMREAD_GRAYSCALE)
```

```
img1=gray_1
img2=gray_2
img2=cv2.resize(img2,(img1.shape[1],img1.shape[0]))
```

✓ 4.1 Arithmetic Operation

✓ 4.1.1 Addition

```
output=cv2.add(img1,img2)

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 2')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(output, cmap='gray')
plt.title('Addition of image 1 and image 2')
plt.axis('off')

plt.show()
```

Image 1



Image 2



Addition of image 1 and image 2



✓ 4.1.2 Subtraction

```
output=cv2.subtract(img1,img2)
```

```
plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 3')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(output, cmap='gray')
plt.title('subtract of image 1 and image 2')
plt.axis('off')

plt.show()
```

Image 1



Image 3



subtract of image 1 and image 2



4.1.3 Multiplication by scaler

```
scaler=1.5
img1_float=img1.astype(np.float32)/255.0
scaled_img=img1_float*scaler
scaled_img=(scaled_img*255).astype(np.uint8)
scaled_img=np.clip(scaled_img,0,255)
scaled_img=scaled_img.astype(np.uint8)

plt.imshow(scaled_img, cmap='gray')
plt.title('Scaled Image')
plt.axis('off')
plt.show()
```

Scaled Image



4.1.4 Multiplication of two images

```
img2_float=img2.astype(np.float32)/255.0
multiplied=img1_float*img2_float
multiplied=(multiplied*255).astype(np.uint8)
multiplied=multiplied.astype(np.uint8)

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
```

```
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 2')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(multiplied, cmap='gray')
plt.title('Multiplication of image 1 and image 2')
plt.axis('off')

plt.show()
```

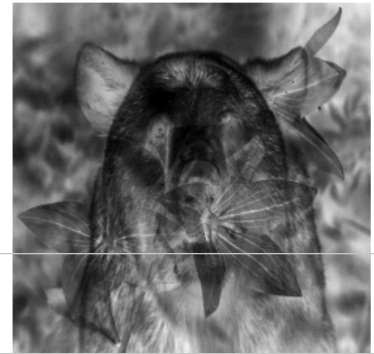
Image 1



Image 2



Multiplication of image 1 and image 2



4.2.6 Division by Scaler

```
divisor=1.5
img1_float=img1.astype(np.float32)/255.0
divided_img=img1_float/divisor
divided_img=(divided_img*255).astype(np.uint8)
divided_img=np.clip(divided_img,0,255)
divided_img=divided_img.astype(np.uint8)

plt.imshow(divided_img, cmap='gray')
plt.title('Divided Image')
plt.axis('off')
plt.show()
```

Divided Image



4.2.6 Division of two images

```
img2_float=img2.astype(np.float32)/255.0

epsilon = 1e-6
divided = img1_float / (img2_float + epsilon)

divided = np.clip(divided, 0, 1)
divided = (divided*255).astype(np.uint8)

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
```

```
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 2')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(divided, cmap='gray')
plt.title('Division of image 1 by image 2')
plt.axis('off')

plt.show()
```

Image 1



Image 2



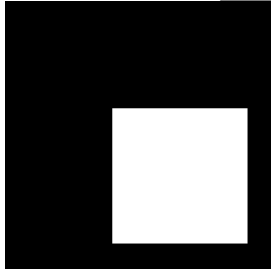
Division of image 1 by image 2



4.2 Logical Operation

```
img1=np.zeros((200,200),dtype=np.uint8)
img2=np.zeros((200,200),dtype=np.uint8)
cv2.rectangle(img1,(30,30),(130,130),255,-1)
cv2.rectangle(img2,(80,80),(180,180),255,-1)
```

ndarray (200, 200) [show data](#)



4.2.1 AND

```
and_result=cv2.bitwise_and(img1,img2)

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 2')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(and_result, cmap='gray')
plt.title('Logical and of image 1 and image 2')
plt.axis('off')

plt.show()
```

Image 1

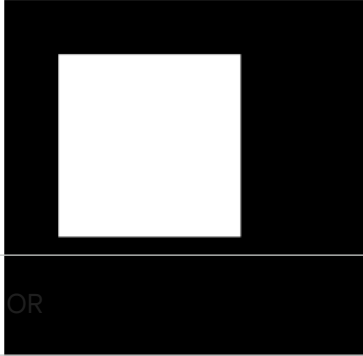
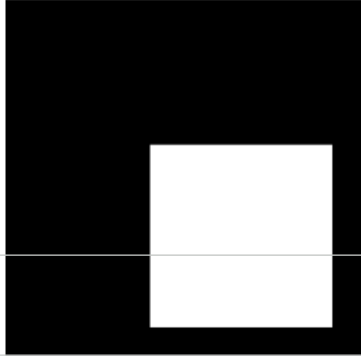
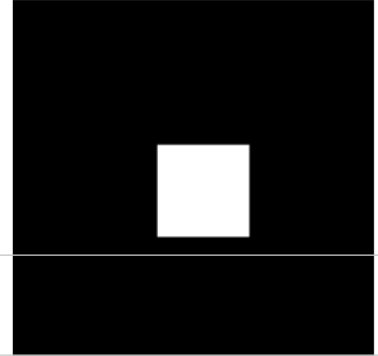


Image 2



Logical and of image 1 and image 2



4.2.2 OR

```
or_result=cv2.bitwise_or(img1,img2)

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 2')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(or_result, cmap='gray')
plt.title('Logical or of image 1 and image 2')
plt.axis('off')

plt.show()
```

Image 1

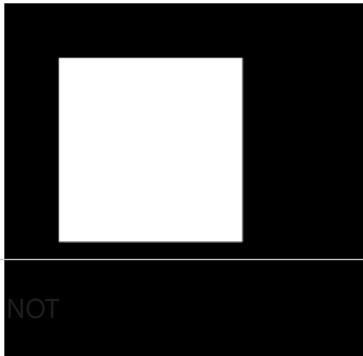
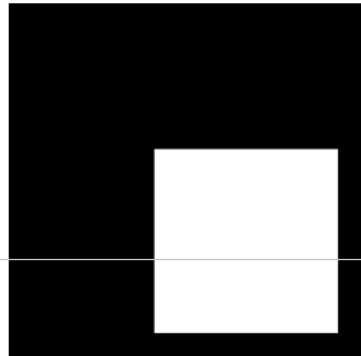
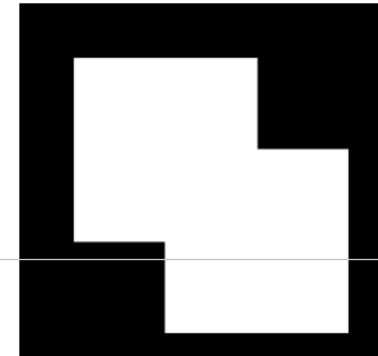


Image 2



Logical or of image 1 and image 2



4.2.3 NOT

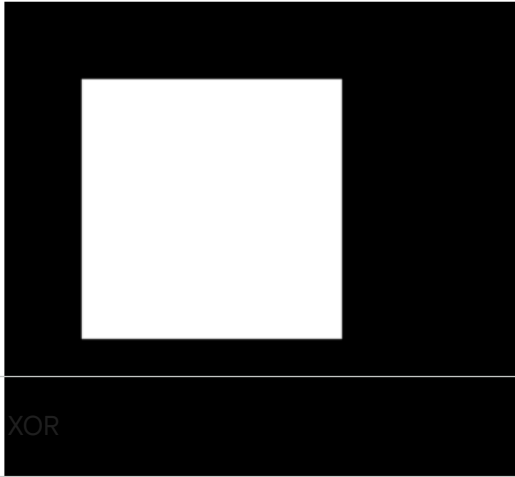
```
not_result=cv2.bitwise_not(img1)

plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(not_result, cmap='gray')
plt.title('Logical not of image 1')
plt.axis('off')

plt.show()
```

Image 1



Logical not of image 1



4.2.4 XOR

```
xor_result=cv2.bitwise_xor(img1,img2)

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 2')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(xor_result, cmap='gray')
plt.title('Logical xor of image 1 and image 2')
plt.axis('off')

plt.show()
```

Image 1

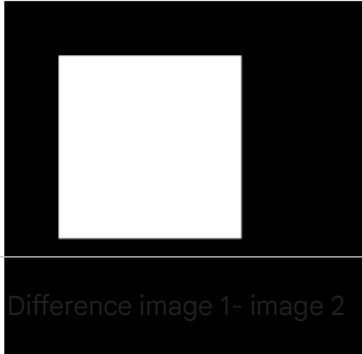
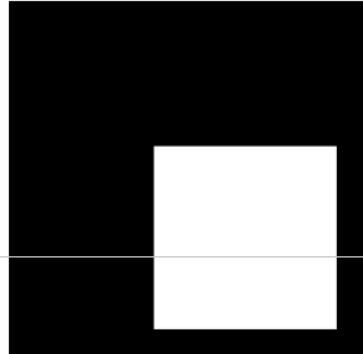
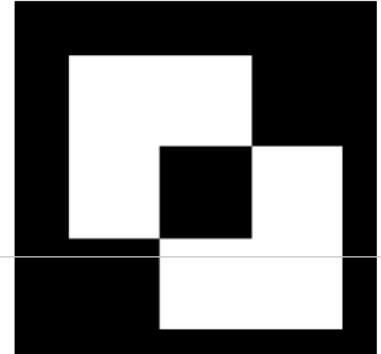


Image 2



Logical xor of image 1 and image 2



4.2.5 Difference image 1- image 2

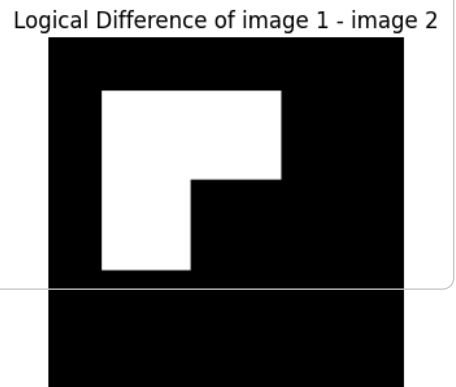
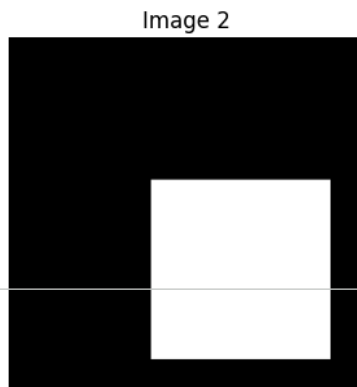
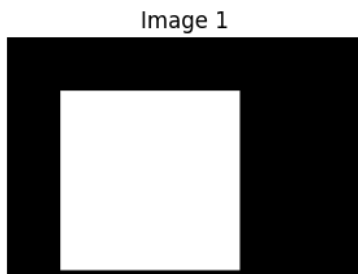
```
difference_result=cv2.bitwise_or(img1,img2) - img2

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 2')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(difference_result, cmap='gray')
plt.title('Logical Difference of image 1 - image 2')
plt.axis('off')

plt.show()
```



4.2.6 Difference image 2 - image 1

```
difference_result=cv2.bitwise_or(img1,img2) - img1

plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(img1, cmap='gray')
plt.title('Image 1')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(img2, cmap='gray')
plt.title('Image 2')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(difference_result, cmap='gray')
plt.title('Logical Difference of image 2 - image 1')
plt.axis('off')

plt.show()
```

Image 1

Image 2

Logical Difference of image 2 - image 1